

Summary

I am a principal software engineer, proficient in C++ and python, and a motivated team player. I enjoy bridging the gap between results obtained by research teams and high quality production code that is integrated into products on various platforms. I also have research experience in Computer Vision, 3D geometry, and Machine Learning.

Professional Experience

Cognex

Fribourg, Switzerland

PRINCIPAL SOFTWARE ENGINEER

August 2022 till now

SENIOR SOFTWARE ENGINEER (C++, COMPUTER VISION)

May 2020 till July 2022

- Objective: Design, implement, test, document, and maintain modular libraries to be used within the Cognex products
- Mission: Implement and optimize new features or algorithms in C++ for multiple platforms (Linux x86_64 or arm64, Windows), design and maintain software architecture and libraries' APIs, manage 3rd party dependencies, modularize and modernize the software stack
- Technologies: C++, Python, Computer Vision, Deep Learning, DevOps (TeamCity CI, Conan, Docker)
- Results: Libraries integrated into desktop, cloud, as well as embedded products. Open source contributions to Torch-TensorRT.

Pix4D

Lausanne, Switzerland

SENIOR R&D ENGINEER

January 2020 till April 2020

SOFTWARE DEVELOPER (C++, COMPUTER VISION)

May 2018 till December 2019

- Objective: Develop core algorithms for a professional drone photogrammetry software suite
- Mission: Design and implement computer vision algorithms, such as deep-learning based image segmentation, multiview stereo, 3D mesh generation, or 3D point cloud filtering with spectral methods, refactor and optimize existing code base and promote good software engineering practices
- Technologies: C++, Python, Photogrammetry, Computer Vision, Machine Learning, Deep Learning, CI/CD (Concourse)
- Results: Adaptation of the main codebase to build on macOS (C++, CMake, Conan) enabling macOS support in all new products. Point cloud spectral filtering algorithms and library integrated into products. PyTorch implementation of a multiview stereo pipeline.

IBM Research

Zürich, Switzerland

POSTDOCTORAL RESEARCHER

October 2017 till April 2018

RESEARCH INTERN

September 2015 to February 2016

- Objective: Automatically extract knowledge from different types of document
- Mission: Conduct research in Image analysis and machine learning (e.g. relational learning), develop and test production-ready code, supervise PhD students and interns
- Technologies: C++, Python, Image analysis, Machine Learning
- Results: Technology transfer to Watson Health, US Patent application, conference article submission

Ecole Polytechnique Fédérale de Lausanne (EPFL)

Lausanne, Switzerland

RESEARCH ASSISTANT

September 2012 to August 2017

- Objective: Automatically predict difficulty of intubation and develop a new 3D face model
- Mission: Conduct research in collaboration with CHUV and nViso, collect data in hospitals, develop a C++ library for facial images analysis, record and align a 3D database of faces, supervise students in facial images analysis projects
- Technologies: C++, Python, Machine Learning, 3D Geometry, Spectral Mesh Processing, 3D Face Models
- Results: EU Patent application, scientific publications

Education

Ph.D., Electrical Engineering

Lausanne, Switzerland

ECOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE (EPFL)

August 2017

- Thesis: *Towards 3D facial morphometry: facial image analysis applications in anesthesiology and 3D spectral nonrigid registration*
- Adviser: Prof. Jean-Philippe Thiran

- Thesis Topic: Thesis Topic: *Difficult Intubation Assessment from Video*
- Area of Study: Major in **information technologies** and minor in **biomedical technologies**

Skills

C++ 11/14/17

>10y., main development language since 2013. Good knowledge of PyTorch, Torch-TensorRT, (modern) CMake, Conan package manager



Python

>5y. NumPy, PyTorch, and Scikit-learn libraries



French

Mother tongue



English

Excellent knowledge, professional language since 2010



Swedish

Good knowledge, exchange year in Sweden, 2002-2003



German

School knowledge, 9 years courses



Selected Publications

Refereed Journal Publications

- **G. L. Cuendet**, P. Schoettker, A. Yüce, M. Sorci, H. Gao, C. Perruchoud, and J.-P. Thiran. Facial image analysis for fully automatic prediction of difficult endotracheal intubation. *IEEE Transactions on Biomedical Engineering*, vol. 63, pp. 328-339, Feb. 2016. doi:10.1109/TBME.2015.2457032
- A. Yüce, H. Gao, **G. L. Cuendet**, J.-P. Thiran. Action Units and Their Cross-Correlations for Prediction of Cognitive Load during Driving. *IEEE Transactions on Affective Computing*, Jun. 2016. doi:10.1109/TAFFC.2016.2584042

Patents

- P. Schoettker, **G. L. Cuendet**, C. Perruchoud, M. Sorci and J.-P. Thiran. Difficult intubation or ventilation prediction system. Patent pending at the European Patent Office, October 2013.

A complete list of publications can be found on <https://gcuendet.github.io/publications/>

Extracurricular

Certificat amateur de violon (certificate of violin amateur studies)

Fribourg, Switzerland

CONSERVATOIRE DE FRIBOURG

June 2009

Chamber music

2009 to present

- Violinist of the "Chromatique" piano trio from 2011 to 2015. We performed public concerts in the French speaking part of Switzerland, playing the classical and romantic repertoire.
- Chamber music master classes in Switzerland and Germany with amongst others: the Mandelring quartet, Paul Cocker, Joel Marosi or the Trio Lenitas.

Sailing

2013 to present

- Club regatta on Surprise from 2015 to 2017, in Lausanne.
- Sailing regularly on Swiss lakes and the Mediterranean Sea.

Chess

2024 to present

- Club player in Fribourg CE.

References

Available upon request